ZIC1-Related Craniosynostosis (Craniosynostosis Type 6, CRS6): A Comprehensive Disease Report

Disease: ZIC1-Related Craniosynostosis **Causal gene: ZIC1** (Zinc finger of the cerebellum 1 / Zic family member 1) **MONDO:** MONDO:0014705 · **OMIM:** 616602 · **Synonyms:** craniosynostosis type 6; ZIC1 craniosynostosis

Summary

ZIC1-Related Craniosynostosis (craniosynostosis type 6, CRS6; OMIM 616602; MONDO:0014705) is a rare autosomal-dominant Mendelian disorder caused by heterozygous **gain-of-function variants in the final (third) exon of the *ZIC1* gene** (chromosome 3q24; HGNC:12872), which encodes a C2H2 zinc-finger transcription factor central to neural development and neural-crest specification. The defining clinical picture is **severe craniosynostosis, specifically involving the coronal sutures (usually bicoronal), together with variable learning disability**, often accompanied by facial dysmorphism and structural brain abnormalities. The disease was first delineated by Twigg and colleagues in 2015 in five families, and the phenotypic spectrum was substantially expanded by Watts and colleagues in 2026 across 22 families / 30 individuals.

The molecular logic of ZIC1 disorders is strikingly dependent on **where in the gene the variant falls**. Truncating (and one missense) variants clustered in the terminal exon 3 **escape nonsense-mediated decay (NMD)**, producing a stable, abnormally active protein that acts through a **gain-of-function / dominant mechanism** — this causes craniosynostosis. In contrast, missense substitutions in exons 1–2 that disrupt the zinc-finger DNA-binding domain cause **loss of ZIC1 function**, producing a **neurodevelopmental disorder without craniosynostosis**; and heterozygous loss of ZIC1 together with its neighbour ZIC4 causes **Dandy-Walker malformation** (cerebellar vermis hypoplasia). ZIC1 therefore sits at the centre of an allelic series in which gain and loss of function produce opposite ends of a developmental spectrum.

Mechanistically, ZIC1 is a **core neural-plate-border / neural-crest specifier** (acting with PAX3) and patterns the **coronal suture via the supraorbital regulatory center**, the boundary between neural-crest-derived frontal bone and mesoderm-derived parietal bone. Gain-of-function ZIC1 drives **osteogenic differentiation (RUNX2, osteocalcin) and Hedgehog signaling (Patched1)** at the expense of other lineages, providing a plausible route to premature bone formation at the suture. ZIC1 integrates **Sonic hedgehog, retinoic acid, and TGFβ/BMP** signaling and regulates engrailed (EN1/EN2) target genes. Management is **surgical** (cranial vault remodeling, fronto-orbital advancement, spring-assisted or distraction techniques) with developmental, ophthalmologic and neurosurgical surveillance; there is no gene-targeted therapy.

Key Findings

Finding 1 — CRS6 is caused by heterozygous gain-of-function variants in the final exon of ZIC1

The foundational study by **Twigg et al. (2015)** identified heterozygous mutations in the **final (third) exon of *ZIC1*** across five families — four nonsense changes and one missense change — producing a "distinct phenotype in which severe craniosynostosis, specifically involving the coronal sutures, and variable learning disability are the most characteristic features" PMID: 26340333. Critically, the **position** of the nonsense mutations within the terminal exon predicts that mutant transcripts **escape nonsense-mediated decay (NMD)** — a prediction the authors confirmed in a cell line derived from an affected individual: *"The location of the nonsense mutations predicts escape of mutant ZIC1 transcripts from nonsense-mediated decay, which was confirmed in a cell line from an affected individual."* Because the mutant message is not degraded, a truncated/altered but **stable protein** is produced, and functional assays in *Xenopus* showed the variants **enhance or alter engrailed-2 target gene expression** — behaviour consistent with a **gain-of-function / dominant** mechanism rather than simple haploinsufficiency. This establishes the gene (ZIC1), the variant location (terminal exon 3), the mechanism (NMD escape → stable active protein), and the core phenotype (coronal craniosynostosis + learning disability).

- **HGNC:** 12872 (ZIC1) · **NCBI Gene:** 7545 · **OMIM gene:** 600470 · **Variant classes:** nonsense (stop-gain), missense, frameshift (all truncating/terminal-exon)
- **Functional consequence:** gain of function / dominant (not loss of function)

Finding 2 & 6 — Genotype–phenotype correlation: exon-3 truncations → craniosynostosis; exon 1–2 missense → NDD alone

The largest cohort to date, **Watts et al. (2026)**, reported *"18 different heterozygous ZIC1 variants from 22 families, comprising 30 individuals"* PMID: 42028696, and resolved a clean genotype–phenotype dichotomy:

Variant location	Variant type	Mechanism	Phenotype
Final exon 3	Truncating (nonsense/ frameshift), rare missense	**Gain of function** (escapes NMD)	Craniosynostosis + facial dysmorphism + structural brain abnormalities + developmental delay (12 families)
Exons 1–2 (zinc-finger domain)	Missense	**Loss of function**	Neurodevelopmental disorder **without** craniosynostosis (10 families)

In the authors' words: *"Variants associated with craniosynostosis were clustered in the final exon (3) and were predominantly truncating variants predicted to escape nonsense-mediated decay, whereas 10 families had a neurodevelopmental disorder alone without craniosynostosis."* This finding is central to clinical interpretation: the same gene produces two distinct disorders depending on variant position and mechanistic direction. It also means that **ACMG classification and phenotype prediction must consider variant location**, not merely predicted protein truncation.

Finding 3 — Mechanism: ZIC1 patterns the coronal suture via the supraorbital regulatory center and controls osteogenic lineage

Twigg et al. localized ZIC1 function anatomically: *"Analysis of mouse embryos revealed a localized domain of Zic1 expression at embryonic days 11.5-12.5 in a region overlapping the supraorbital regulatory center, which patterns the coronal suture"* PMID: 26340333. The supraorbital regulatory center is the osteogenic/suture-patterning hub from which the frontal and parietal bones and the intervening coronal suture arise. Independently, **Thottappillil et al. (2025)** showed that ZIC1 overexpression in human skeletal progenitor cells *"led to an increase in osteoblast antigen expression (RUNX2, OCN), activation of Hedgehog signaling (Patched1), and an increase in brown adipogenesis markers"* PMID: 40151950 — i.e., excess ZIC1 **drives osteogenic differentiation and Hedgehog signaling**, a direct mechanistic route to

premature suture ossification. The coronal suture is a developmental boundary between **neural-crest-derived frontal bone and paraxial-mesoderm-derived parietal bone**, whose integrity depends on Shh-responsive supraorbital cells PMID: 26589924.

Finding 7 — ZIC1 is a core neural-plate-border / neural-crest specifier (with PAX3)

ZIC1 operates upstream in craniofacial development as a **neural-crest specifier**. Bellchambers et al. state that *"The zinc finger of the cerebellum 1 (ZIC1) transcription factor is considered a core component of the vertebrate gene regulatory network that specifies neural crest fate at the neural plate border"* PMID: 34638777. In *Xenopus*, zic1 and pax3 are the core neural-plate-border factors initiating the neural-crest gene regulatory network, and their relative levels differentially regulate downstream NC genes (snai2, sox8): *"These differences correlate with dynamic expression of the neural plate border factors pax3 and zic1"* PMID: 41718037. Because the **frontal bone derives from cranial neural crest** and the **parietal bone from paraxial mesoderm**, ZIC1's role in NC specification mechanistically links it to the craniofacial skeleton and the coronal suture PMID: 32916911.

Finding 8 — ZIC1 integrates Shh, retinoic acid, and TGFβ/BMP signaling

A review by **Aruga & Millen (2018)** describes Zic-mediated developmental control operating through regulation of neuronal progenitor proliferation/differentiation and patterning of the cerebellar primordium, with *"embryonic interactions with sonic hedgehog signaling, retinoic acid signaling, and TGFβ signaling ... described during mouse cerebellar development"* PMID: 29442326. The same review independently confirms that *"ZIC1 gain of function mutations contribute to coronal craniosynostosis, a rare skull malformation."* These three pathways (Shh, RA, TGFβ/BMP) are the principal signaling systems ZIC1 interfaces with and are relevant to both the osteogenic/suture branch and the cerebellar branch of ZIC1 biology.

Finding 4 — Allelic opposite: ZIC1/ZIC4 loss of function causes Dandy-Walker malformation

Loss-of-function biology sits at the other end of the ZIC1 spectrum. **Grinberg et al. (2004)** established that heterozygous deletion of the linked genes ZIC1 and ZIC4 underlies Dandy-Walker malformation (DWM), and that *"Mice with a heterozygous deletion of these two linked genes have a phenotype that closely resembles DWM, providing a mouse model for this malformation"* PMID: 15338008. **Blank et al. (2011)** showed the cerebellar hypoplasia results from reduced postnatal granule-cell progenitor proliferation via Shh-dependent function:

"Expression of the Shh-downstream genes Ptch1, Gli1 and Mycn was downregulated in Zic1/4 mutants" PMID: 21307096. This LoF mechanism (reduced Hedgehog output) is the mechanistic mirror image of the GoF craniosynostosis mechanism (increased Hedgehog/osteogenic output), tying both ends of the spectrum to the same Shh pathway.

Finding 5 — Variable expressivity, postnatal suture fusion, and parental gonadal mosaicism

Vandervore et al. (2018) described two siblings with a novel heterozygous frameshift at the end of ZIC1 exon 3 presenting with neonatal microcephaly, agenesis of the corpus callosum, brachycephaly, reduced posterior fossa volume, cerebellar/pons hypoplasia, scoliosis, tethered cord and partial rhombencephalosynapsis. The variant was absent in both parents: *"The mutation was not detected in DNA of both parents, which suggests parental gonadal mosaicism"* PMID: 30391508 — a finding with direct recurrence-risk implications. Importantly, *"Craniosynostosis was not present at birth, which explains why ZIC1 mutations were not initially considered,"* demonstrating **postnatal onset of suture fusion** and variable expressivity. A separate report (**Murakami et al. 2024**) found that a de novo 3q inversion with a breakpoint **7 kb downstream of ZIC1** produced a bicoronal craniosynostosis / rhombencephalosynapsis phenotype via transcriptional dysregulation of ZIC1 — showing that regulatory (position-effect) disruption can phenocopy coding gain-of-function PMID: 37950019.

Section-by-Section Report

1. Disease Information

ZIC1-Related Craniosynostosis (CRS6) is a rare autosomal-dominant Mendelian skull malformation syndrome in which **premature fusion of the coronal sutures** (craniosynostosis) co-occurs with **variable learning disability / developmental delay**, frequently with facial dysmorphism and structural brain abnormalities. The **causal gene is ZIC1**, encoding the zinc-finger-of-the-cerebellum-1 transcription factor.

- **Key identifiers:** OMIM 616602 (phenotype); MONDO:0014705; HGNC:12872 / NCBI Gene 7545 / OMIM gene 600470 (ZIC1). ICD-10 Q75.0 (craniosynostosis); ICD-11 LD24.0. MeSH: "Craniosynostoses" (D003398). Orphanet lists coronal/complex craniosynostosis entities; a ZIC1-specific Orphanet code was not confirmed in this investigation.

- **Synonyms:** craniosynostosis type 6; ZIC1 craniosynostosis; craniosynostosis caused by mutation in ZIC1.
- **Information source:** Disease-level knowledge is derived from **aggregated case series and cohort studies** (Twigg 2015, 5 families; Watts 2026, 22 families/30 individuals; Vandervore 2018, 1 family), not EHR/population registries, reflecting the disorder's rarity.

2. Etiology

- **Primary cause — genetic:** heterozygous **gain-of-function variants in the final (third) exon of ZIC1** that escape NMD, yielding a stable, abnormally active protein PMID: 26340333. No environmental or infectious cause is implicated.
- **Genetic risk factors:** the causal variant itself (exon-3 truncating/missense). No modifier genes or susceptibility loci are established for CRS6; penetrance for craniosynostosis among exon-3 variant carriers appears high but expressivity is variable.
- **Environmental / protective / gene-environment factors: None established.** CRS6 is a monogenic disorder; there is no evidence of environmental risk factors, protective alleles, lifestyle factors, or gene–environment interactions. (Not applicable for this Mendelian disorder.)

3. Phenotypes

Phenotype	Type	HPO suggestion	Onset / severity / frequency
Coronal craniosynostosis (usually bicoronal)	Physical/ structural	HP:0004440 (coronal craniosynostosis); HP:0001363 (craniosynostosis)	Congenital or **postnatal** onset; severe; near-defining in the CS subgroup
Learning disability / developmental delay	Behavioral/ cognitive	HP:0001328 (specific learning disability); HP:0001263 (global developmental delay)	Childhood; variable severity; common
Facial dysmorphism	Clinical sign	HP:0001999 (abnormal facial shape)	Congenital; variable; frequent in CS subgroup
Structural brain abnormalities (callosal agenesis, cerebellar/pons hypoplasia, rhombencephalosynapsis)	Imaging/ structural	HP:0001274 (agenesis of corpus callosum); HP:0001321 (cerebellar hypoplasia)	Congenital; variable
Microcephaly	Physical	HP:0000252	Neonatal (reported)
Scoliosis; tethered cord	Skeletal/ neural	HP:0002650 (scoliosis); HP:0002144 (tethered cord)	Childhood; reported in sib pair
Reduced posterior fossa volume	Imaging	HP:0002707	Congenital

Evidence: coronal synostosis + learning disability as the characteristic dyad PMID: 26340333; facial dysmorphism + structural brain abnormalities + developmental delay in the craniosynostosis subgroup PMID: 42028696; microcephaly, callosal agenesis, cerebellar/pons hypoplasia, scoliosis, tethered cord, rhombencephalosynapsis PMID: 30391508.

Quality-of-life impact: craniosynostosis carries risk of raised intracranial pressure and ophthalmologic sequelae requiring surgery and surveillance; learning disability affects education/independence. Disorder-specific QoL instruments for CRS6 were not identified. Non-syndromic single-suture data show unicoronal synostosis is associated with ophthalmologic complications (~23.6%) and that surgical timing affects ICP/hydrocephalus risk PMID: 42153715 — relevant context, though not CRS6-specific.

4. Genetic / Molecular Information

- **Causal gene: ZIC1** (HGNC:12872; NCBI Gene 7545; chromosome 3q24; OMIM gene 600470).
- **Pathogenic variants (craniosynostosis):** nonsense, frameshift and rare missense changes **clustered in terminal exon 3**, predicted to **escape NMD** → stable protein → **gain of function** PMID: 26340333; PMID: 42028696. A frameshift at the end of exon 3 was documented with gonadal mosaicism PMID: 30391508.
- **Pathogenic variants (NDD without craniosynostosis): missense in exons 1–2** disrupting the zinc-finger domain → **loss of function** PMID: 42028696.
- **Classification:** per ACMG/AMP, exon-3 truncating variants escaping NMD are classified as pathogenic/likely pathogenic in the appropriate phenotypic context; **specific HGVS nomenclature was intentionally deferred to ClinVar/HGMD** during the investigation to avoid transcription error.
- **Allele frequency:** these are rare/private disease variants, not present at appreciable frequency in gnomAD (consistent with a severe dominant disorder); ZIC1 is highly constrained.
- **Origin:** germline (de novo or inherited); **gonadal/germline mosaicism documented** PMID: 30391508. No somatic disease role.
- **Chromosomal abnormalities:** a de novo **3q inversion with breakpoint 7 kb downstream of ZIC1** caused a ZIC1-dysregulation phenotype (bicoronal craniosynostosis + rhombencephalosynapsis), implicating position/TAD effects PMID: 37950019. Heterozygous **ZIC1;ZIC4 deletions** cause the allelic Dandy-Walker malformation PMID: 15338008.
- **Modifier / epigenetic:** no established modifier genes; variable expressivity suggests unidentified modifiers or stochastic/developmental factors. No disease-specific epigenetic signature reported.

5. Environmental Information

Not applicable. CRS6 is a monogenic gain-of-function disorder with no identified environmental, lifestyle, toxic, or infectious contributing factors.

6. Mechanism / Pathophysiology

Ordered causal chain (gain-of-function / craniosynostosis branch):

1. A heterozygous truncating (or rare missense) variant arises in the **final exon 3 of ZIC1** (germline; sometimes de novo or via gonadal mosaicism). *(demonstrated)*
2. Because the premature stop/frameshift lies in the terminal exon, the mutant transcript **escapes nonsense-mediated decay**, so a **stable, truncated/altered ZIC1 protein is produced**. *(demonstrated — confirmed in patient cell line)* [PMID: 26340333]
3. The stable mutant protein acts in a **gain-of-function / dominant** manner, **altering/ enhancing ZIC1 target-gene regulation** (e.g., engrailed-2/EN targets in Xenopus assay). *(demonstrated in vitro)*
4. In the developing skull, excess/aberrant ZIC1 activity in the **supraorbital regulatory center** (where Zic1 is expressed at mouse E11.5–12.5) **dysregulates patterning of the coronal suture**. *(inferred from expression domain + functional data)* [PMID: 26340333]
5. Elevated ZIC1 activity **drives osteogenic differentiation (RUNX2, osteocalcin) and activates Hedgehog signaling (Patched1)** in skeletal progenitors, **at the expense of alternative lineages**. *(demonstrated in human progenitor overexpression; mechanistic bridge to suture)* [PMID: 40151950]
6. This **tips the osteogenic/non-osteogenic boundary at the neural-crest/mesoderm interface** toward premature bone formation, **leading to premature fusion of the coronal suture (craniosynostosis)**. *(inferred, consistent with coronal-boundary biology)* [PMID: 26589924;

   P 32916911]
7. Premature coronal fusion **results in abnormal skull shape and can lead to raised intracranial pressure, ophthalmologic sequelae, and contributes (with ZIC1's brain-patterning roles) to learning disability / structural brain abnormalities**. *(clinical observation)* [PMID: 26340333;

   P 42028696]

Branch (loss-of-function): Missense in exons 1–2 → disrupted zinc-finger DNA binding → **loss of ZIC1 function → neurodevelopmental disorder without craniosynostosis** [PMID: 42028696]; and ZIC1/ZIC4 deletion → **reduced Shh-dependent granule-cell proliferation (↓Ptch1, Gli1, Mycn)** → cerebellar hypoplasia / **Dandy-Walker malformation** [PMID: 21307096;

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P 15338008

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                ZIC1 locus (3q24)
                       |
     +-----------------+------------------+
 EXON 3 truncating                EXON 1-2 missense /
 (escapes NMD)                    ZIC1;ZIC4 deletion
     | GAIN of function               | LOSS of function
     v                                v
up osteogenesis (RUNX2/OCN)     down Shh output (Ptch1/Gli1/Mycn)
up Hedgehog (Patched1)          down granule-cell proliferation
     v                                v
premature CORONAL suture fusion  cerebellar hypoplasia
+ learning disability             -> Dandy-Walker / NDD (no CS)
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Pathways / processes / ontology suggestions: - Molecular pathways: **Sonic hedgehog**, **retinoic acid**, **TGFβ/BMP** signaling; engrailed (EN1/EN2) target regulation [PMID: 29442326;

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- Biological processes (GO): regulation of transcription by RNA polymerase II (GO:0006357); neural crest cell fate specification (GO:0014036); cranial suture morphogenesis; ossification (GO:0001503); osteoblast differentiation (GO:0001649); smoothened/Hedgehog signaling pathway (GO:0007224); regulation of neuron differentiation (GO:0045664). - Cell types (CL): cranial neural crest cell (CL:0000333 neural crest cell); osteoblast (CL:0000062); skeletal/osteogenic progenitor; cerebellar granule cell (CL:0001031) for the LoF branch. - Subcellular (GO CC): nucleus (GO:0005634) — ZIC1 is a nuclear transcription factor. - Chemical entities (CHEBI): retinoic acid (CHEBI:50211); zinc (CHEBI:29105, zinc-finger cofactor).

7. Anatomical Structures Affected

- **Organ level:** skull/calvaria (primary); brain (secondary — cerebellum, corpus callosum, pons, posterior fossa); orbit/visual system; vertebral column.
- **Body systems:** skeletal and nervous systems predominantly.

- **Tissue/cell:** cranial **neural-crest-derived frontal bone** and **mesoderm-derived parietal bone**; the **coronal suture** at their boundary; osteogenic progenitors; cerebellar granule neurons (LoF branch).
- **UBERON suggestions:** cranial suture / coronal suture (UBERON:0000063); frontal bone (UBERON:0001678); parietal bone (UBERON:0001679); cerebellum (UBERON:0002037); corpus callosum (UBERON:0002336); posterior cranial fossa; skull (UBERON:0003129).
- **Lateralization:** craniosynostosis is typically **bilateral/bicoronal**; can be asymmetric.

8. Temporal Development

- **Onset:** congenital developmental disorder; **craniosynostosis may be present at birth or develop postnatally** — in at least one family it was NOT present at birth PMID: 30391508. Brain/structural anomalies are congenital.
- **Progression:** suture fusion is **progressive during infancy/early childhood**; skull deformity and risk of raised ICP evolve until surgically corrected. Developmental/learning features are generally **stable/chronic** (non-degenerative).
- **Critical period:** the therapeutic window for cranial vault surgery is infancy/early childhood; surgical timing influences ICP and hydrocephalus risk (general craniosynostosis data) PMID: 42153715.
- **Duration:** lifelong condition; no spontaneous remission of the genetic defect.

9. Inheritance and Population

- **Inheritance: autosomal dominant** (heterozygous variants; de novo or inherited).
- **Penetrance / expressivity:** high penetrance for craniosynostosis among exon-3 variant carriers but **variable expressivity** (range of brain/skeletal involvement, and postnatal vs congenital synostosis) PMID: 30391508.
- **Mosaicism: parental gonadal (germline) mosaicism documented**, relevant to recurrence-risk counseling PMID: 30391508.
- **Epidemiology: ultra-rare** — fewer than ~30 families reported worldwide at the time of the Watts 2026 cohort (22 families/30 individuals) PMID: 42028696. Precise prevalence/incidence are not established; ZIC1 accounts for a small minority of craniosynostosis cases. No founder effect, consanguinity role, sex bias, or ethnic predilection established.

10. Diagnostics

- **Clinical/imaging:** diagnosis of craniosynostosis by **skull CT (gold standard for suture fusion)**, 3D CT, and clinical exam; brain MRI for structural anomalies (callosal agenesis, cerebellar/pons hypoplasia, rhombencephalosynapsis, posterior fossa volume) PMID: 30391508. Ophthalmologic monitoring (including OCT of the optic nerve, feasible in children with craniosynostosis PMID: 41735680) for papilledema/raised ICP.
- **Genetic testing (confirmatory): single-gene sequencing of ZIC1, craniosynostosis gene panels, whole-exome or whole-genome sequencing**. WGS / long-read sequencing adds value for **structural/regulatory variants** — a de novo 3q inversion downstream of ZIC1 was resolved only by **Nanopore long-read sequencing** PMID: 37950019. Interpretation must weigh **variant location (exon 3 vs 1–2)** and NMD-escape prediction.
- **Differential diagnosis:** other syndromic coronal craniosynostoses — **Saethre-Chotzen (TWIST1, TCF12)**, Muenke (FGFR3), Apert/Crouzon (FGFR2), and craniosynostosis with MSX2 gain of function; also Gomez-Lopez-Hernandez syndrome (clinical overlap with the ZIC1 rhombencephalosynapsis presentation) PMID: 37950019. TWIST1/TCF12/EFNA4/EphA4/Jagged1 act in the same coronal-suture boundary network [PMID: 34878091; P 16540516; P 19201948; P 20727876].
- **Screening:** no population newborn screening; **cascade/familial testing** and genetic counseling for at-risk relatives are appropriate given dominant inheritance and documented mosaicism.

11. Outcome / Prognosis

- **Survival:** not a life-limiting genetic defect per se; **life expectancy is not clearly shortened** and the disorder is chronic rather than fatal. No disease-specific mortality data for CRS6.
- **Morbidity:** driven by **raised intracranial pressure, visual compromise, neurodevelopmental/learning disability, and need for (often multiple) cranial surgeries**. Comparative syndromic craniosynostosis experience (e.g., Apert syndrome 53-year follow-up) suggests many patients have near-normal lifespan without high cancer risk PMID: 42082645 — reassuring context, though not ZIC1-specific.

- **Prognostic factors:** severity of synostosis, presence/severity of brain malformation, degree of learning disability, and timeliness of surgical intervention. Surgical timing influences ICP/hydrocephalus outcomes in craniosynostosis generally PMID: 42153715.

12. Treatment

No gene-targeted or pharmacologic disease-modifying therapy exists. Management is **surgical and supportive**:

- **Surgical (primary):** cranial vault remodeling, **fronto-orbital advancement and remodeling**, **endoscope-assisted suturectomy with postoperative orthotic/helmet therapy**, **spring-assisted cranioplasty**, and **distraction osteogenesis**; vector/technique chosen by phenotype and age PMID: 42686138; PMID: 42853819. NCIT suggestions: Cranial Vault Remodeling; Fronto-Orbital Advancement; Craniectomy.
- **Supportive/rehabilitative:** developmental support, physical/occupational/speech therapy for developmental delay (benefit of early intensive rehabilitation illustrated in a related syndromic craniosynostosis case PMID: 41893374); ophthalmologic and neurosurgical surveillance for ICP.
- **Experimental/targeted:** none established; the gain-of-function mechanism theoretically suggests allele-specific knockdown (ASO/siRNA) as a conceptual avenue, but **no clinical trials exist for ZIC1 craniosynostosis**.

13. Prevention

- **Primary prevention:** not possible for a monogenic de novo/inherited disorder. **Genetic counseling** is the main preventive tool, including discussion of **gonadal mosaicism and recurrence risk** PMID: 30391508.
- **Reproductive options:** prenatal testing / preimplantation genetic testing for families with a known pathogenic ZIC1 variant.
- **Secondary/tertiary prevention:** early imaging detection of synostosis and ICP; timely surgery; ophthalmologic and developmental surveillance to prevent complications.

14. Other Species / Natural Disease

- **Orthologues:** mouse *Zic1* (NCBI Gene 22771), with linked *Zic4*; *Xenopus zic1*; zebrafish zic1. Strongly conserved across vertebrates.
- **Model-organism disease:** mouse *Zic1/Zic4* loss recapitulates **Dandy-Walker malformation** (the LoF allelic disorder), and Zic1 loss causes cerebellar hypoplasia and vertebral defects

[PMID: 15338008;

P 21307096

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No naturally occurring companion-animal "ZIC1 craniosynostosis" is documented (OMIA entry not identified). No zoonotic/transmission relevance (genetic disorder).

- **Comparative biology:** evolutionary conservation of ZIC1 as a neural-crest/neural-plate-border specifier underlies the cross-species utility of *Xenopus* and mouse models [PMID: 34638777;

P 41718037

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15. Model Organisms

Model	Type	Findings / recapitulation	PMID
Mouse *Zic1/Zic4* heterozygous deletion	Mammalian, LoF	Closely resembles Dandy-Walker malformation (cerebellar size/ foliation defects)	15338008; 21307096
Mouse embryo Zic1 expression mapping	Mammalian	Zic1 domain overlaps supraorbital regulatory center patterning the coronal suture (E11.5–12.5)	26340333
Xenopus engrailed-2 / NC-GRN assays	Amphibian, in vivo functional	GoF variants alter/enhance engrailed-2 targets; zic1/pax3 regulate NC specification	26340333; 41718037
Human skeletal progenitor cells (in vitro)	Cellular	ZIC1 overexpression → ↑RUNX2/OCN, ↑Hedgehog (Patched1), brown adipogenesis	40151950

Limitation of models: no published mouse model reproduces the **gain-of-function exon-3 ZIC1 craniosynostosis** phenotype directly; current mammalian models capture the **loss-of-function (Dandy-Walker)** end, so a dedicated knock-in model of a terminal-exon truncation is a key gap.

Mechanistic Model / Interpretation

ZIC1 disorders are best understood as a **dosage- and location-dependent allelic series around a single transcription factor** that controls both neural-crest specification and Hedgehog-linked osteogenesis. The unifying variable is the **direction and magnitude of ZIC1 activity**:

- **Too much / aberrant activity** (terminal-exon truncations escaping NMD) → excess osteogenic + Hedgehog drive in the supraorbital/coronal region → **premature coronal fusion (craniosynostosis)** plus brain-patterning disturbance → learning disability.
- **Too little activity** (zinc-finger missense; ZIC1/ZIC4 deletion) → reduced Shh-dependent granule-cell proliferation → **cerebellar hypoplasia / Dandy-Walker**, or NDD without craniosynostosis.

The **Hedgehog pathway is the shared mechanistic hub**: gain of function **raises** Hedgehog output (Patched1 up in osteogenic cells), while loss of function **lowers** it (Ptch1/Gli1/Mycn down in cerebellum). This mirror symmetry, plus ZIC1's position at the neural-crest/mesoderm boundary that defines the coronal suture, explains why coronal sutures are specifically vulnerable and why the same gene yields opposite malformations.

Evidence Base

PMID	Study	Role in this report
26340333	Twigg et al. 2015 — *Gain-of-function ZIC1 mutations in coronal craniosynostosis*	Foundational: gene, exon-3 location, NMD escape, GoF mechanism, supraorbital expression
42028696	Watts et al. 2026 — *Expanding the ZIC1 phenotypic spectrum*	Largest cohort; genotype–phenotype (CS vs NDD) correlation
40151950	Thottappillil et al. 2025 — *ZIC1 overexpression & osteogenic/brown-adipogenic differentiation*	Mechanism: ZIC1 drives osteogenesis + Hedgehog
30391508	Vandervore et al. 2018 — *ZIC1 sibling phenotype*	Gonadal mosaicism, postnatal synostosis, variable expressivity
37950019	Murakami et al. 2024 — *3q inversion dysregulating ZIC1*	Regulatory/position-effect phenocopy; long-read sequencing utility
21307096	Blank et al. 2011 — *Zic1/4 loss & Dandy-Walker*	LoF branch; Shh target downregulation
15338008	Grinberg et al. 2004 — *ZIC1;ZIC4 deletion in DWM*	LoF allelic disorder; mouse model
34638777	Bellchambers et al. 2021 — *SUMOylation & ZIC neural-crest specification*	ZIC1 as core neural-crest specifier
41718037	Montequin & LaBonne 2026 — *NPB TF dynamics*	zic1/pax3 at neural plate border
29442326	Aruga & Millen 2018 — *ZIC1 review*	Shh/RA/TGFβ integration; independent GoF-craniosynostosis confirmation
26589924 / 32916911	Ishii 2015 / Siismets & Hatch 2020	Coronal-suture developmental biology; NC/mesoderm boundary

Limitations and Knowledge Gaps

1. **Ultra-rare disease, small N.** The core phenotype rests on ~27 families total across the key reports; frequency/penetrance estimates are imprecise.
2. **No GoF animal model.** Existing mammalian models capture the loss-of-function (Dandy-Walker) end; no knock-in reproduces exon-3 truncation craniosynostosis, so the GoF causal chain (steps 4–6) is partly **inferred**.
3. **Variant-level detail deferred.** Specific HGVS nomenclature and per-variant ACMG classifications were intentionally not transcribed here (to avoid error) and should be drawn from ClinVar/HGMD.
4. **Mechanistic bridge is correlative.** The link from "ZIC1 drives osteogenesis/Hedgehog in vitro" to "premature coronal fusion in vivo" is biologically plausible but not directly demonstrated in a craniosynostosis model.
5. **No epidemiology, QoL, or natural-history registries** specific to CRS6; prognosis is extrapolated from the broader craniosynostosis literature.
6. **No disease-specific epigenetic, metabolomic, or proteomic data.**

Proposed Follow-up Experiments / Actions

1. **Generate a conditional knock-in mouse** carrying a terminal-exon ZIC1 truncation (NMD-escape) to test whether it reproduces coronal synostosis and to map osteogenic/Hedgehog dysregulation in vivo.
2. **Single-cell / spatial transcriptomics of the embryonic coronal suture** in GoF models to define the cell populations (CL terms) and Hedgehog-pathway state at the neural-crest/mesoderm boundary.
3. **Curate a definitive ZIC1 variant table** from ClinVar/HGMD stratified by exon and mechanism, with ACMG classifications, to operationalize the genotype–phenotype rule for diagnostic labs.
4. **Assess Hedgehog-pathway modulation** (e.g., smoothened antagonists) in GoF osteogenic-progenitor models as a proof-of-concept for mechanism-based intervention.
5. **Establish a CRS6 natural-history/QoL registry** capturing age at synostosis onset, ICP events, surgeries, and visual/neurodevelopmental outcomes.
6. **Prospectively apply long-read genome sequencing** in ZIC1-negative but clinically compatible cases to detect regulatory/structural variants (as in the 3q inversion case).

Evidence source types: human clinical (Twigg 2015, Watts 2026, Vandervore 2018, Murakami 2024); model organism (Grinberg 2004, Blank 2011 mouse; Montequin 2026 Xenopus); in vitro (Thottappillil 2025 human progenitors); review/computational (Aruga & Millen 2018, Bellchambers 2021).